

Introduction to Social Influence Models

Further issues in ALAAM analysis)

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Preamble

- All material is on the workshop repository
<https://github.com/johankoskinen/ALAAM>
 - ▶ Download the RMarkdown file CHDH-SNA-2.Rmd
 - ▶ Download the (proto) manual https://github.com/johankoskinen/ALAAM/blob/main/alaam_effects.pdf
- In order to run the Markdown you need
 - ▶ The R-package 
 - ▶ The RStudio interface  RStudio
- We will predominantly use the packages
 - ▶ sna
 - ▶ network
- as well as **balaam.R** from GitHub



Agenda

- 1 Model selection
- 2 Missing data
- 3 Interactions
 - SBC
- 4 Fully Bayesian
- 5 HALAAM
- 6 MALAAM
- 7 Further topics



Model selection

Picking the 'best' model



Posterior deviance

The deviance is defined as minus twice the log likelihood

$$D(\boldsymbol{\theta}) = -2 \log[p_{\boldsymbol{\theta}}(\mathbf{y}|\mathbf{X})].$$

Aitkin et al. (2017) graphical comparison of models can be done through comparing the posterior distribution of the deviance

Assume a sample

$$\boldsymbol{\theta}_0, \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_T$$

Calculate the deviance $D(\boldsymbol{\theta}_t)$ for the parameters in your posterior.



Posterior deviance: important

We *cannot* evaluate log likelihood

$$p_{\theta}(\mathbf{y}|\mathbf{X}),$$

because of $\psi(\theta)$.

But for pairs $\tilde{\theta}$ and θ , we can approximate $\hat{\lambda}(\theta, \tilde{\theta}) \approx \exp\{\psi(\theta) - \psi(\tilde{\theta})\}$.

Intuition: for bridges $\tilde{\theta} = \theta^{(0)}, \theta^{(1)}, \dots, \theta^{(M)} = \theta$, we draw

$$\mathbf{y}_0^{(j)}, \mathbf{y}_{2k}^{(j)}, \dots, \mathbf{y}_{3k}^{(j)}, \mathbf{y}_{4k}^{(j)}, \dots, \mathbf{y}_{Tk}^{(j)} \sim p_{\theta^{(j)}}(\mathbf{y} | \mathbf{X})$$

and use¹ $\bar{\mathbf{z}}^{(j)} = \frac{1}{T} \sum \mathbf{z}(\mathbf{y}_t^{(j)}, \mathbf{X})$ to get estimate

$$\hat{\lambda}(\theta, \tilde{\theta}) \approx \exp\{\psi(\theta) - \psi(\tilde{\theta})\}$$

NB: Sensitive to T and thinning k - samples $\{\mathbf{y}_t^{(j)}\}$ have to be good

¹Requires a bit more thought ...

Deviance information criterion

Using the posterior distribution of the deviance, we can calculate

$$DIC = E[D(\boldsymbol{\theta})] + V(D(\boldsymbol{\theta}))/2$$

Models with **smaller** DIC preferred to models with **LARGER** DIC



Example: s50

Example (Modern contraceptive use in rural Kenya $n = 1303$)

In CHDH-SNA-2.Rmd

```
my.dev.post <- post.deviance.alaam(res.DC.2,# the estimation object
                                   numBridges=5,# the safer default is 20
                                   thinning.like = 1000,# thinning in drawing y; should be
                                   sample.size = 20,# number of y drawn for each bridge
                                   cov.sample.burnin = NULL,
                                   printFreq=10,# print to screen after done 10
                                   mult.fact = 30,# depreciated
                                   num.outs=100)# number of devaiance evaluations to return
```

Where res.DC.2 comes from call to estimate.alaam

Example: s50

Example (Smoking in s50)

In CHDH-SNA-2.Rmd deviance computed as

```
my.dev.post <- post.deviance.alaam(res.DC.2,# the estimation object
                                     numBridges=5,# the safer default is 20
                                     thinning.like = 1000,# thinning in drawing y; should be 1
                                     sample.size = 20,# number of y drawn for each bridge
                                     cov.sample.burnin = NULL,
                                     printFreq=10,# print to screen after done 10
                                     mult.fact = 30,# depreciated
                                     num.outs=100)# number of devaiance evaluations to
```

Where res.DC.2 comes from call to estimate.alaam

Example: s50

Example (Smoking in s50)

In CHDH-SNA-2.Rmd, plotting deviances

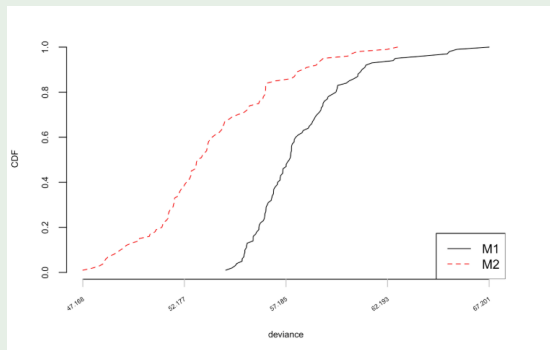
```
plot.deviance.alaam(dev.1=ind.post.dev, dev.2=my.dev.post)
```

Where ind.post.dev and my.dev.post comes from call to
post.deviance.alaam

Example: s50

Example (Smoking in s50)

In CHDH-SNA-2.Rmd, plotting deviances



And DIC

```
c( alaam.dic(ind.post.dev), alaam.dic(my.dev.post) )
```

Missing data

Missing outcomes



Missing data (cp Bayesian data augmentation for ERGM)

Under assumption of Missing at Random (MAR)

Define the missing data mechanism $f(I|y, \phi)$, where

$$I_i = \begin{cases} 1, & \text{if response } y_i \text{ is unobserved for } i \\ 0, & \text{else} \end{cases}$$

update (impute) missing response by toggling and accepting w.p.

$$\min \left[1, \exp\{\theta^\top (z(\Delta_i y, x) - z(y, x))\} \frac{f(I|\Delta_i y, \phi)}{f(I|y, \phi)} \right]$$

where $\Delta_i y$ is y with element i toggled and set to $1 - y_i$.

Update ϕ , with MH-updating and Hastings ratio

$$\min \left\{ 1, \frac{f(I|y, \phi^*)\pi(\phi^*)}{f(I|y, \phi)\pi(\phi)} \right\}.$$



Missing data (cp Bayesian data augmentation for ERGM)

In the actual estimation, simply define

$$y_i = \begin{cases} 1, & \text{if response } y_i = 1 \text{ is unobserved for } i \\ 0, & \text{if response } y_i = 0 \text{ is unobserved for } i \\ NA, & \text{if response is missing for } i \end{cases}$$

Sampling will return draws

$$(\theta^{(0)}, \mathbf{y}_{miss}^{(0)}), (\theta^{(1)}, \mathbf{y}_{miss}^{(1)}), \dots, (\theta^{(M)}, \mathbf{y}_{miss}^{(M)})$$



Example: ALAAM with social influence (1)

Example (Modern contraceptive use in rural Kenya $n = 1303$)

Using balaam

```
res.DC.0 <- estimate.alaam( mcUse ~ odegree
                             + Age
                             + Female
                             + HasChildren
                             + relevanOthersApprove
                             + relevanOthersUse
                             + mcUseConflict
                             + simple,
                             village , adjacency = adj,
                             Iterations = 3000)
```

Example: ALAAM with social influence (2)

Example (Modern contraceptive use in rural Kenya $n = 1303$)

balaam reports $Y_i = NA$ for two obs

Network is directed with 1303 nodes

```
[1] "you have 2 missing entries in response"
```

```
NULL
```

```
[1] "you have p: 9"
```

observed stats: 459 332 536 14419 255 373 342 376 287

number of covariates: 7

A thinning of 1 ,

(parameter) burn-in of 1 , and

iterations of 3000 ,

yields a sample size of 3000 .

Example: ALAAM with social influence (3)

Example (Modern contraceptive use in rural Kenya $n = 1303$)

Each iteration m imputes for $y_{miss}^{(m)}$

you have done 300 iterations out of 3000

theta: -0.921 0.482 0.089 -0.045 -1.123 1.652 1.335 0.314 -0.418

observed stats: 459 332 536 14419 255 373 342 376 287

imputed ones: 0 out of 2

you have done 600 iterations out of 3000

theta: -0.314 0.48 -0.105 -0.049 -1.243 1.883 1.293 0.056 -0.385

observed stats: 460 332 536 14474 255 374 343 376 288

imputed ones: 1 out of 2

Example: ALAAM with social influence (4)

Example (Modern contraceptive use in rural Kenya $n = 1303$)

Each iteration m imputes for $y_{miss}^{(m)}$

you have done 300 iterations out of 3000

theta: -0.921 0.482 0.089 -0.045 -1.123 1.652 1.335 0.314 -0.418

observed stats: 459 332 536 14419 255 373 342 376 287

imputed ones: 0 out of 2

you have done 600 iterations out of 3000

theta: -0.314 0.48 -0.105 -0.049 -1.243 1.883 1.293 0.056 -0.385

observed stats: 460 332 536 14474 255 374 343 376 288

imputed ones: 1 out of 2

Example: ALAAM with social influence (5)

Example (Modern contraceptive use in rural Kenya $n = 1303$)

At total of K imputations $\mathbf{y}_{miss}^{(k_1)}, \dots, \mathbf{y}_{miss}^{(k_K)}$ are stored

```
> names(res.DC.0)
```

```
[1] "Thetas"      "ALAAMobj"     "ResTab"      "priorSigma"  "priorMu"
```

```
[6] "imputed.obs" "silent"       "do.scaling"  "PropSigma"   "MPLE"
```

```
[11] "saveFreq"    "missFreq"     "initTheta"   "acc.ratio"
```

```
> dim(res.DC.0$imputed.obs)
```

```
[1] 1303 100
```

Because missFreq=10 in call to estimate.alaam

You can set this!

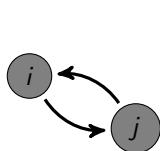
Interactions with contagion

More complicated contagion effects

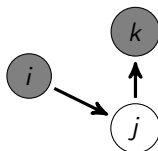


More elaborate effects

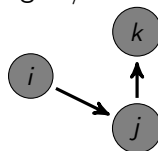
A number of more elaborate forms of contagion/influence are admissible



reciprocal



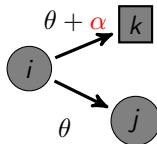
indirect



reinforced indir

Interaction:

influence from some nodes can be θ and for others $\theta + \alpha$



SBC (Koskinen and Daraganova, 2022)

Stockholm Birth Cohort (SBC) cohort study, Stockholm Metropolitan area (Stenberg et al., 2006; Stenberg et al. 2007).

- best-friend network with a cap of three nominations (May 1966)
- Let y be indicators $y_i = 1$ of whether pupils i said that they **intended to proceed to higher secondary school**, and $y_i = 0$ otherwise (see Koskinen and Stenberg, 2012)
- Here: 19 school classes, six of which are from a school in a suburb in the south of Stockholm and the rest are from three inner-city schools
- The proportion of **missing** entries range from 0 to 0.286



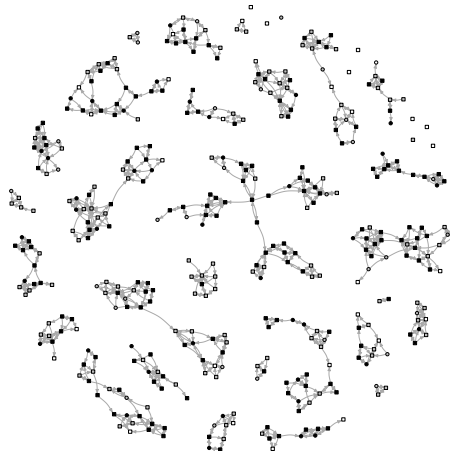


Figure: Bffs in 4 schools. Squares (girl) and circles (boys), and outcome black ($y_i = 1$),

More elaborate effects - interaction example

Example (Simple contagion of intention to go to higher secondary school)

	mean	sd	ESS	SACF 10	SACF 30	2.5 perc	97.5 perc
intercept	-9.67	1.11	178.03	0.68	0.32	-11.83	-7.51
contagion	0.16	0.10	183.10	0.68	0.32	-0.04	0.35
indegree	-0.07	0.11	183.55	0.67	0.32	-0.29	0.13
sex	-0.09	0.29	134.35	0.70	0.39	-0.66	0.47
family attitude	0.48	0.09	164.22	0.70	0.32	0.33	0.65
marks	0.99	0.15	168.66	0.68	0.32	0.69	1.28
social class 1	0.59	0.32	198.40	0.66	0.24	-0.06	1.19

Table: Posterior summaries for model with controls estimated for contagion-model for progression to upper-secondary school in SBC (thinned sample of 10,000 iterations, taking every 20th iteration, with burnin of 1000)

More elaborate effects - interaction example

Example (Contextual contagion of intention to go to higher secondary school)

	mean	sd	ESS	SACF 10	SACF 30	2.5 perc	97.5 perc
intercept	-10.13	1.19	168.32	0.76	0.44	-12.81	-8.04
contagion	0.24	0.12	143.31	0.72	0.39	0.02	0.48
indegree	-0.08	0.12	122.80	0.75	0.41	-0.33	0.13
sex	-0.09	0.28	126.04	0.76	0.45	-0.69	0.47
family attitude	0.48	0.08	140.26	0.72	0.38	0.34	0.65
marks	1.01	0.14	265.08	0.72	0.40	0.76	1.31
composition	0.91	0.55	137.33	0.74	0.39	-0.25	1.97
social class 1	0.57	0.34	143.59	0.73	0.37	-0.07	1.21
contagion int	-0.21	0.16	152.15	0.72	0.37	-0.51	0.11

Table: Posterior summaries for model with controls estimated for contagion-model for progression to upper-secondary school in SBC (thinned sample of 10,000 iterations, taking every 20th iteration, with burnin of 1000) **with social class interacted with contagion**

Example: s50

Example (Smoking in s50)

In CHDH-SNA-2.Rmd we try

```
res.DC.ADC.1 <- estimate.alaam(smoke ~ odegree
                                +alcohol
                                +sport
                                +simple
                                +alcohol*simple,
                                my.data, adjacency=adj,
                                Iterations=5100,
                                par.burnin=100,
                                thinning=10)
```

Where alcohol*simple is interaction



Fully Bayesian

Specifying proper priors



Fully Bayesian: what priors?

Assuming

$$\mathbf{y} \mid \mathbf{X} \sim \text{ALAAM}(\boldsymbol{\theta}, \mathbf{X})$$

The model is

$$P(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) = \exp\{\boldsymbol{\theta}^\top \mathbf{z}(\mathbf{y}, \mathbf{X}) - \psi(\boldsymbol{\theta})\}$$

The aim of the Bayesian inference scheme is to obtain the posterior

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}) \propto P(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta})\pi(\boldsymbol{\theta})$$

where

$$\pi(\boldsymbol{\theta})$$

is the **prior distribution** for the parameters that *quantify our uncertainty about the parameter values prior to observing data*.



Fully Bayesian: what priors?

How can a human *quantify their uncertainty about the parameter values prior to observing data?*

For ALAAM this is really hard! Possible choices

- Default prior: constant $\pi(\boldsymbol{\theta}) \propto 1$
- Convenient: Multivariate normal distribution $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
 - ▶ Diagonal: $\boldsymbol{\Sigma} = \lambda \mathbf{I}$
 - ▶ Scaling: $\boldsymbol{\Sigma} = \lambda(\mathbf{X}^\top \mathbf{X})^{-1}$
- Caution: setting $\boldsymbol{\mu} = \mathbf{0}$ pulls posteriors towards 0 - bad for e.g. intercept if $\bar{y}$ small
- Experimental: use a prior to 'fix' a nuisance parameter



Example: s50

Example (Smoking in s50)

In CHDH-SNA-2.Rmd we try

```
priorSigma <- diag(c(1,.2,.1,1,.0001,.01))  
priorMu <- matrix(0,6,1)
```

and use `priorSigma` as argument `priorSigma = priorSigma` in call to `estimate.alaam`

Hierarchical ALAAM

A multilevel version of ALAAM

In stand-alone CHDH-HALAAM.Rmd



Hierarchical ALAAM: preamble

The routines for analysing outcomes

$$\mathbf{y}^{(1)}, \mathbf{y}^{(2)}, \dots, \mathbf{y}^{(G)}$$

for G independently observed networks

$$\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \dots, \mathbf{X}^{(G)}$$

is implemented in *very old* code in `MultivarALAAMalt.R`



Hierarchical ALAAM: principle

Following Koskinen and Snijders' (2023) work on SAOMs, we assume that independently for each group $g = 1, \dots, G$,

$$\mathbf{y}^{(g)} \mid \mathbf{X}^{(g)}, \boldsymbol{\theta}^{(g)} \sim ALAAM(\boldsymbol{\theta}^{(g)}, \mathbf{X}^{(g)})$$

where parameters are partitioned

$$\boldsymbol{\theta}^{[g]} = \begin{pmatrix} \boldsymbol{\gamma}^{[g]} \\ \boldsymbol{\eta} \end{pmatrix},$$

into a set of group-specific parameters $\boldsymbol{\gamma}^{[g]} \in \boldsymbol{\Gamma} \subset \mathbb{R}^q$, and a common parameter $\boldsymbol{\eta} \in \boldsymbol{H} \subset \mathbb{R}^r$, $q + r = p$.



Hierarchical ALAAM: 'fixed' effects

The common parameter, η , can be used to parse out **group-level** effects, e.g. Public/private school, gender composition of school class, etc. We may assume that η , are independent of $\gamma^{[1]}, \gamma^{[2]}, \dots, \gamma^{[G]}$, with prior

$$\pi(\eta \mid \mu_\eta, \Sigma_\eta, \gamma^{[1]}, \gamma^{[2]}, \dots, \gamma^{[G]}) = \pi(\eta \mid \mu_\eta, \Sigma_\eta)$$

which for convenience may be assumed to be $\mathcal{N}_r(\mu_\eta, \Sigma_\eta)$.



Hierarchical ALAAM: 'random' effects

The group-level parameters allow, e.g. the intercept and the contagion effect to vary across groups.

We assume

$$\gamma^{[g]} \stackrel{iid}{\sim} \mathcal{N}_r(\mu_\gamma, \Sigma_\gamma).$$

Assuming that $\gamma^{[g]}$ follow a multivariate normal, it is common (cp Gelman et al., 1995) to assume a Normal-inverse-Wishard prior for the parameters μ_γ and Σ_γ

$$\mu_\gamma \mid \Sigma_\gamma \sim \mathcal{N}_q(\mu_0, \Sigma_\gamma/\kappa_0), \text{ and } \Sigma_\gamma \sim \mathcal{IW}_r(\mathbf{\Lambda}_0, \nu_0)$$



Hierarchical ALAAM: the DAG

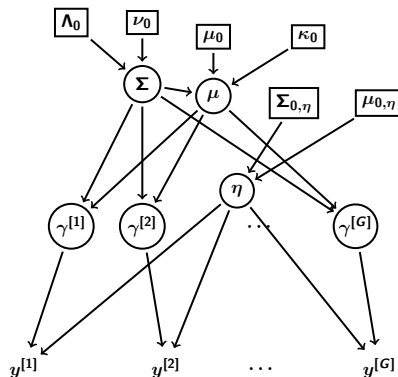
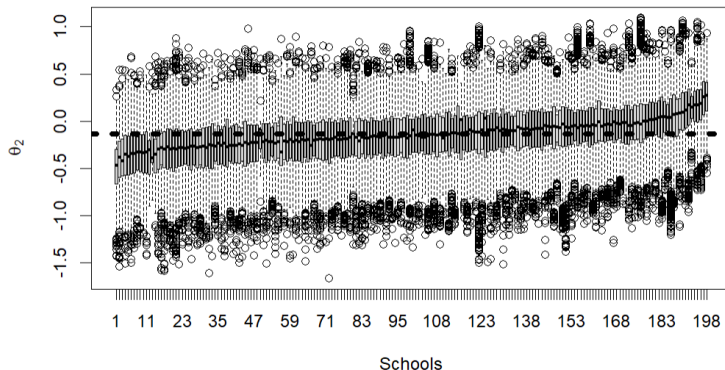


Figure: Dependence structure of hierarchical network model for $\mathbf{Y}^{[g]}$, $g = 1, 2, \dots, G$.

Hierarchical ALAAM: example (1)

Acke Arvidsson (Master's thesis) analysed hundreds of school classes in SBC

Outcome is leader ($y_i = 1$) or not ($y_i = 0$); Network BFF

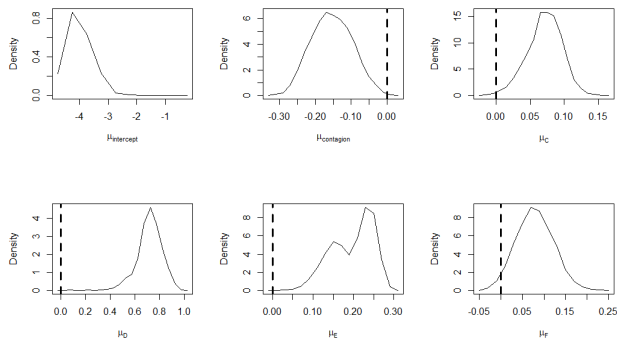


Posteriors for group-level contagion $\gamma_2^{(g)}$

Hierarchical ALAAM: example (2)

Acke Arvidsson (Master's thesis) analysed hundreds of school classes in SBC

Outcome is leader ($y_i = 1$) or not ($y_i = 0$); Network BFF



Posteriors for μ

C: indegree; D: sex; E: average test score; F: communication



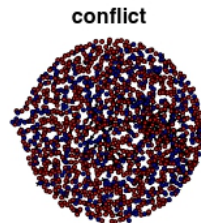
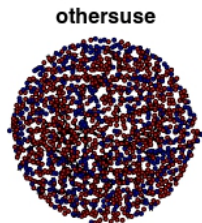
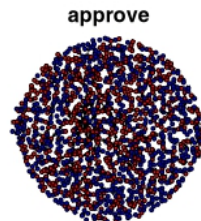
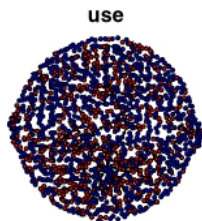
Multivariate ALAAM

A Multivariate version of ALAAM

In stand-alone CHDH-MALAAM.Rmd



Kenya dataset: ALAAM - why **one** dependent variable?



Data structure

For each individual $i \in V$ binary vector

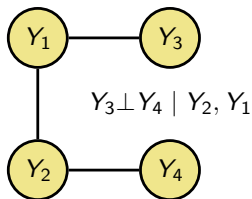
$$\mathbf{y}_i = \begin{bmatrix} Y_{1i} \\ Y_{2i} \\ Y_{3i} \\ Y_{4i} \end{bmatrix}$$

with possible dependencies through network ties $\mathbf{X} = (X_{ij})$



Graphical models: Graphs

Assume $K = 4$ binary variables



$Y_3 \perp Y_4 \mid Y_2, Y_1$

Graph maps out dependence among variables
Cliques are interactions, e.g.

$\{Y_1, Y_3\}$

Absence of edges:
Pairs that are conditionally independent

Think Simpson's paradox

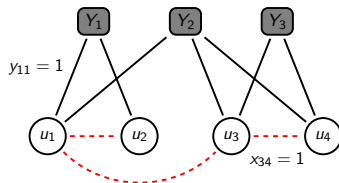
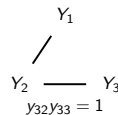
Data structure

(a) *Data matrix*

	Y_1	Y_2	Y_3
u_1	1	1	0
u_2	1	0	0
u_3	0	1	1
u_4	1	0	1

(b) *Communication matrix*

	u_1	u_2	u_3	u_4
u_1	—	1	1	0
u_2	1	—	0	0
u_3	1	0	—	1
u_4	0	0	1	—

(c) *Respondent to respondent/variable ties*(d) *One-mode projection*

People by variables

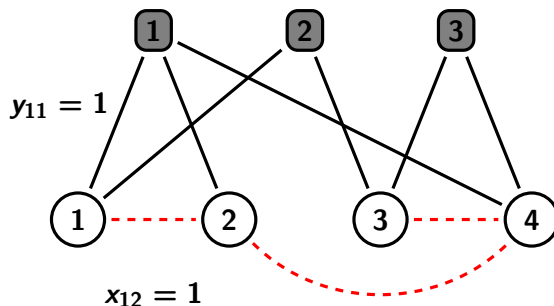


Figure: A two-mode network with $n = 4$ people (round nodes) affiliated with $K = 3$ variables (square, filled nodes)

People by variables: social influence

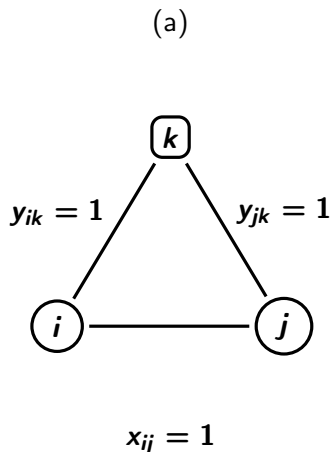


Figure: Evidence for social influence/contagion: people that hang out more likely to exhibit same behaviour/attitudes

People by variables: social influence

We can estimate a (non-homogenous) ERGM for the people by people by variables network



The multivariate ALAAM (the key result)

We may specify a dependence graph (or partial dependence graphs) on $\mathcal{Y} \times \mathcal{X}$, to obtain a joint model

$$p(\mathbf{Y}, \mathbf{X}) = \exp\left\{\underbrace{\theta_{\mathbf{Y}}^{\top} \mathbf{z}_{\mathbf{Y}}(\mathbf{Y})}_{\text{outcome}} + \underbrace{\theta_{\mathbf{X}}^{\top} \mathbf{z}_{\mathbf{X}}(\mathbf{X})}_{\text{network}} + \underbrace{\theta_{\mathbf{YX}}^{\top} \mathbf{z}_{\mathbf{YX}}(\mathbf{Y}, \mathbf{X})}_{\text{interaction}} + \psi_{\mathbf{YX}}(\theta)\right\}.$$

The conditional distribution, is obtained as

$$p(\mathbf{Y} \mid \mathbf{X}) = \exp\{\theta_{\mathbf{Y}}^{\top} \mathbf{z}(\mathbf{Y}) + \theta_{\mathbf{YX}}^{\top} \mathbf{z}(\mathbf{Y}, \mathbf{X}) - \psi_{\mathbf{Y}|\mathbf{X}}(\theta)\}$$

noting that the terms $\theta_{\mathbf{X}}^{\top} \mathbf{z}(\mathbf{X})$ and $\psi_{\mathbf{YX}}(\theta)$ cancel in $p(\mathbf{Y}, \mathbf{X})/p(\mathbf{X})$, and in the denominator, we are left with

$$\sum_{\mathbf{Y} \in \mathcal{Y}} \exp\{\theta_{\mathbf{Y}}^{\top} \mathbf{z}(\mathbf{Y}) + \theta_{\mathbf{YX}}^{\top} \mathbf{z}(\mathbf{Y}, \mathbf{X})\} \equiv e^{\psi_{\mathbf{Y}|\mathbf{X}}(\theta)}$$

ERGO: we can estimate Multivariate ALAAM using ERGM for multilevel networks



Obtaining the standard log-linear, graphical model

Assume that each $i \in V$ independent

$\Rightarrow \theta_{\mathbf{y}\mathbf{x}} = 0,$

parameters for interactions of type $y_{ik}y_{jk}$ are zero, and

Parameters for interactions of type y_{ik} and y_{jk} are equal across $i \in V$
(Homogeneity)

We obtain a **standard graphical model**



Obtaining the standard log-linear, graphical model

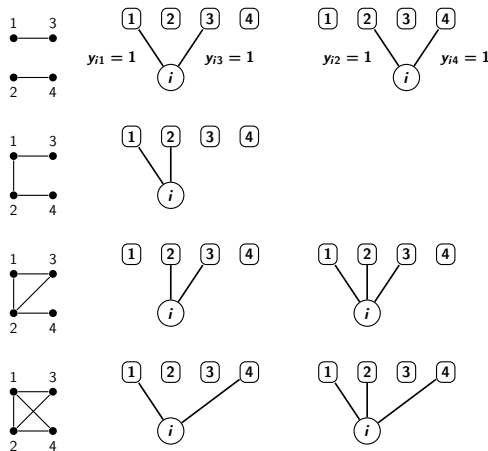


Figure: A sequence of decomposable graphs on $K = 4$ variables that differ in exactly one edge along with the increasing sequence, row-wise, of sufficient ERGM statistics for a two-mode $n \times K$ network

Obtaining a log-linear with peer-dependence

Relax independence across $i \in V$

But disallow social influence (through network)

$\Rightarrow \theta_{\mathbf{y}\mathbf{x}} = 0,$

parameters for interactions of type $y_{ik}y_{jk}$ are non-zero, and

Parameters for interactions of type y_{ik} and y_{jk} are equal across $i \in V$
(Homogeneity)

We obtain a homogenous ERGM for 1×2 -mode graphs



Obtaining a log-linear with peer-dependence (additional stats)

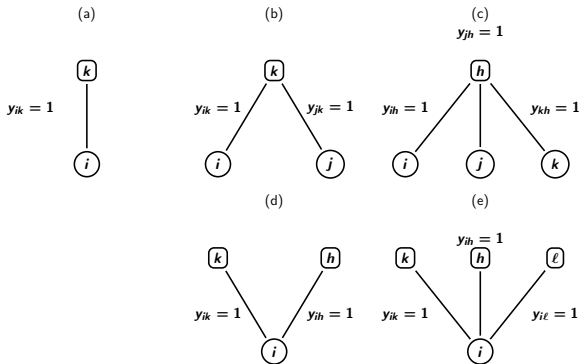


Figure: Sufficient graph statistics under Markov dependence for two-mode networks

Obtaining a log-linear with social influence

If in addition, you **allow** social influence (through network)

$\Rightarrow \theta_{\mathbf{y}\mathbf{x}} \neq 0$,

parameters for interactions of type $y_{ik}y_{jk}$ are **non-zero**, and

Parameters for interactions of type y_{ik} and y_{jk} are equal across $i \in V$
(Homogeneity)



Obtaining a log-linear with social influence

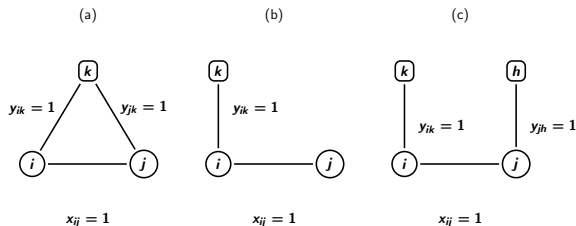


Figure: Social influence (a), two-way interaction associating a variable with node degree (b), and three-way interaction (c)

Results for Kenya data

Conflict (Y_1); Use (Y_2); Approve (Y_3); Others Use (Y_4)

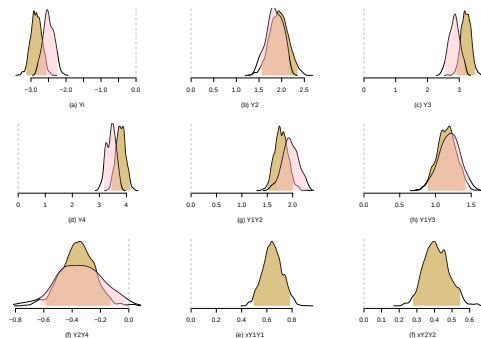


Figure: The posterior distributions for two models with HPD intervals. The posteriors for the independent model (pink) and the social influence model (dark golden)

Results for Kenya data

- If you think MC use causes *conflict*, you are more likely to *use*, and you are more likely to think that it is important that others *approve*.
- if you think that it is important that *others* use you are less likely to *use*
- ✓ if your friends think that use causes *conflict* you are more likely to think that use causes conflict
- ✓ if your friends use you are more likely to *use*



Model selection

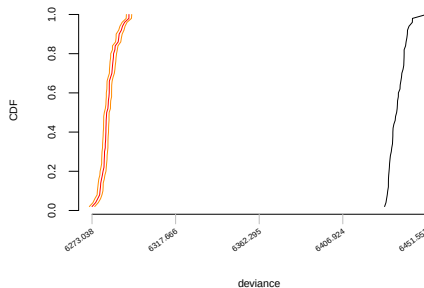


Figure: The posterior deviances (Aitkin et al., 2017) $-2\ell(\theta; \mathbf{Y})$ for $\theta \sim \pi(\theta | \mathbf{Y})$ for two models.

Further topics

Further complications



Further topics: topics

- Missing **NOT** at random
- Missing network ties
- Marginal effects: Titanic
- Multivariate ALAAM
- Snowball sample or outlier nodes can change

